

Hydrocarbons

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Alkanes ($C_n H_{2n+2}$) contain only single C-C bonds, are relatively unreactive, and undergo free radical substitution reactions (e.g., halogenation).
- Alkenes ($C_n H_{2n}$) contain at least one C=C double bond; they are more reactive than alkanes and undergo addition reactions (e.g., with H_2 , X_2 , HX , H_2O).
- Alkynes ($C_n H_{2n-2}$) contain at least one C triple-bond C; they undergo addition reactions similar to alkenes, but can add two equivalents of reagent.
- Markovnikov's rule: in the addition of HX to an unsymmetrical alkene, H adds to the carbon with more hydrogens already attached (the more stable carbocation forms).
- Anti-Markovnikov addition (peroxide effect) occurs specifically with HBr in the presence of peroxides, proceeding via a free radical mechanism instead.
- Aromatic hydrocarbons (e.g., benzene) have a stable, delocalized pi electron system (6 pi electrons, following Huckel's $4n+2$ rule) and undergo electrophilic substitution reactions rather than addition.
- Combustion of hydrocarbons produces CO_2 and H_2O (complete combustion) or CO /soot (incomplete combustion), releasing energy.



Alkene C=C



Alkyne C≡C

Alkane, alkene, and alkyne: single, double, and triple carbon-carbon bonds.

2. Formula Sheet

Alkane general formula	$C_n H_{(2n+2)}$
Alkene general formula	$C_n H_{2n}$
Alkyne general formula	$C_n H_{(2n-2)}$
Huckel's rule (aromaticity)	Number of pi electrons = $4n + 2$
Markovnikov's rule (qualitative)	H adds to carbon with more H atoms already
Combustion of alkane (general)	$C_n H_{(2n+2)} + O_2 \rightarrow CO_2 + H_2O$
Ozonolysis (qualitative)	$C=C + O_3 \rightarrow$ two carbonyl compounds
Degree of unsaturation	$DoU = (2C+2-H)/2$ (for hydrocarbons, no N or halogens)

3. Practice Problems (22 questions)

1. Write the general formula for alkanes and find the molecular formula of the alkane with 5 carbon atoms.
2. State the type of reaction alkanes typically undergo with halogens in the presence of sunlight.
3. Apply Markovnikov's rule to predict the major product when HBr adds to propene ($CH_3-CH=CH_2$).
4. State the peroxide effect (anti-Markovnikov addition) and the specific reagent for which it applies.
5. Find the degree of unsaturation of a hydrocarbon with molecular formula C_5H_8 .
6. State Huckel's rule for aromaticity and verify that benzene (6 pi electrons) satisfies it.
7. Explain why alkenes are more reactive than alkanes toward addition reactions.
8. State the products of complete combustion of methane (CH_4).
9. Explain why benzene undergoes electrophilic substitution rather than addition reactions, despite having double bonds.
10. Find the number of moles of H_2 required to completely hydrogenate 1 mole of an alkyne to an alkane.
11. State the mechanism type (free radical, electrophilic, or nucleophilic) for the halogenation of alkanes.
12. Explain the difference in reactivity between alkanes, alkenes, and alkynes in general terms.
13. Find the molecular formula of the alkyne with 4 carbon atoms.
14. State what is meant by 'ozonolysis' and its use in determining the position of a double bond in an alkene.
15. Explain why cycloalkanes have the same general formula as alkenes ($C_n H_{2n}$).
16. State the conditions required for the free radical halogenation of an alkane (e.g., methane + Cl_2).
17. Explain Markovnikov's rule using carbocation stability as the underlying reason.
18. Find the number of sigma bonds in ethane (C_2H_6).

19. State one major industrial or everyday use of alkenes (e.g., ethylene for polyethylene).
20. Explain why aromatic compounds like benzene are unusually stable compared to a hypothetical non-aromatic triene.
21. Find the degree of unsaturation for benzene (C₆H₆) and interpret its meaning (rings + pi bonds).
22. State the difference in the type of addition reaction alkynes undergo compared to alkenes (in terms of equivalents of reagent).

Answer Key

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| 1. C _n H _{2n+2} ; for n=5, C ₅ H ₁₂ | 2. Free radical substitution |
| 3. Major product: 2-bromopropane (CH ₃ -CHBr-CH ₃), Br adds to the more substituted carbon per Markovnikov's rule | 4. Conceptual - HBr adds anti-Markovnikov (H to more substituted carbon) in the presence of peroxides, via free radical mechanism |
| 5. DoU = 2 (e.g., a ring + a double bond, or two double bonds, or a triple bond) | 6. Conceptual - Huckel's rule: 4n+2 pi electrons for aromaticity; benzene has 6 = 4(1)+2, satisfying n=1 |
| 7. Conceptual - the pi bond in alkenes is weaker and more accessible, making it more reactive toward electrophiles/addition | 8. CO ₂ + H ₂ O (complete combustion) |
| 9. Conceptual - addition would destroy the stable delocalized aromatic pi system; substitution preserves aromaticity, which is energetically favored | 10. 2 moles of H ₂ |
| 11. Free radical mechanism (initiation, propagation, termination steps) | 12. Conceptual - alkynes > alkenes > alkanes in general reactivity toward addition, due to increasing pi bond character/electron density |
| 13. C ₄ H ₆ | 14. Conceptual - ozonolysis cleaves the C=C bond into two carbonyl compounds, revealing the double bond's original position |
| 15. Conceptual - cycloalkanes have one ring instead of one degree of unsaturation from a double bond, giving the same H-deficiency as alkenes (same general formula) | 16. Conceptual - requires UV light or heat to initiate homolytic Cl-Cl bond cleavage, generating radicals |
| 17. Conceptual - the more stable (more substituted) carbocation intermediate forms preferentially, directing where H adds | 18. 7 sigma bonds (1 C-C + 6 C-H) |
| 19. Conceptual - e.g., ethylene used to manufacture polyethylene plastic | 20. Conceptual - delocalization of the 6 pi electrons over the ring (resonance) provides extra stabilization energy compared to a non-aromatic triene with localized double bonds |
| 21. DoU = 4 (3 pi bonds + 1 ring, consistent with the benzene ring structure) | 22. Conceptual - alkynes can undergo two successive additions (adding 2 equivalents of reagent) since they have two pi bonds, compared to one addition for alkenes |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.